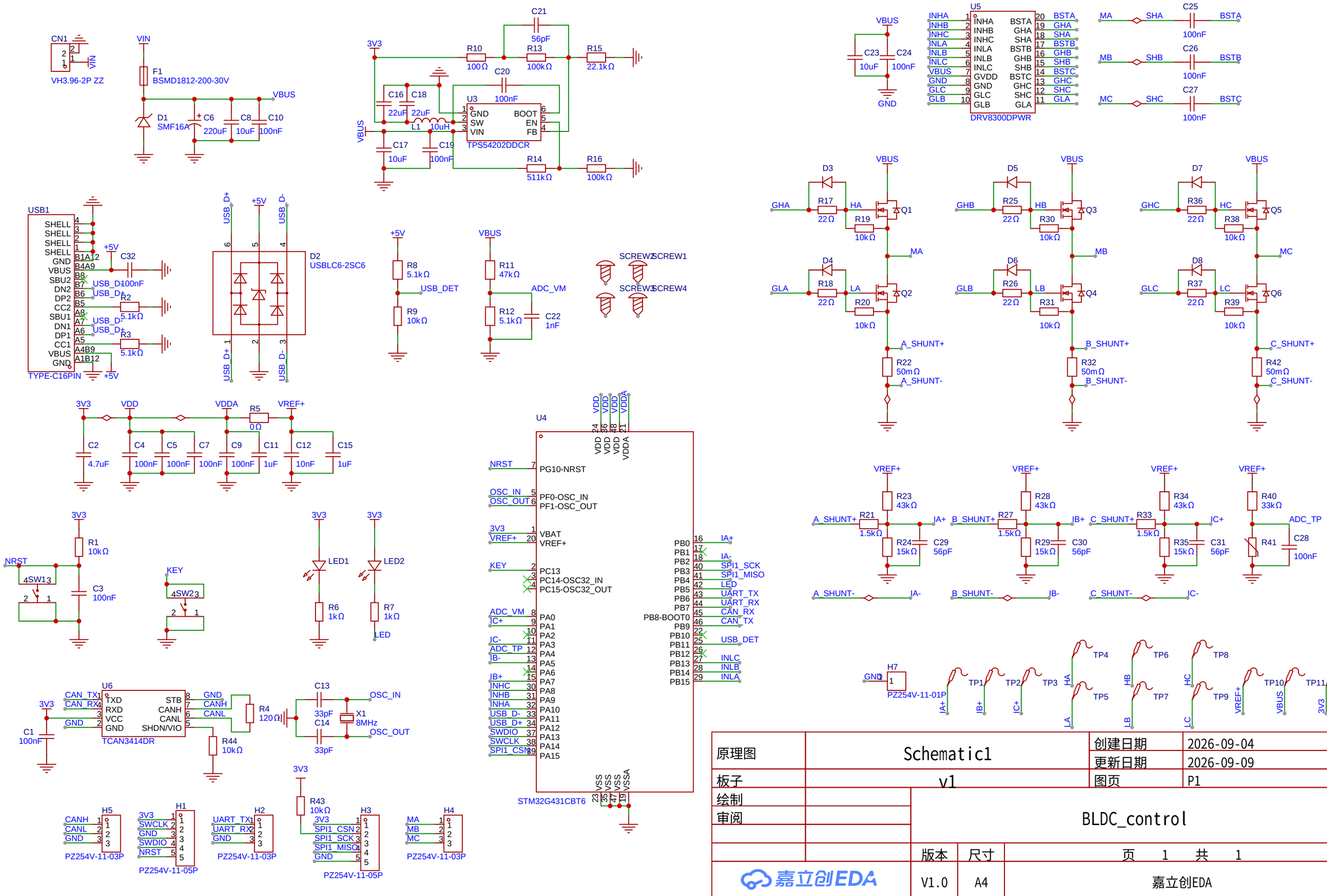

CN1
VH3.96-2P ZZ
VIN
F1
BSMD1812-200-30V
D1
SMF16A
C6
220uF
C8
10uF
C10
100nF
VBUS
3V3
C21
56pF
R10
100Ω
R13
100kΩ
R15
22.1kΩ
C20
100nF
C16
22uF
C18
22uF
U3
TPS54202DDCR
GND
SW
VIN
BOOT
EN
FB
L1
10uH
C17
10uF
C19
100nF
R14
511kΩ
R16
100kΩ
C23
10uF
C24
100nF
GND
U5
DRV8300DPWR
INHA
INHB
INHC
INLA
INLB
INLC
VBUS
GND
GLC
GLB
GVDD
SHA
SHB
SHC
GHA
GHB
GHC
GLA
GLB
GLC
MA
SHA
C25
100nF
BSTA
MB
SHB
C26
100nF
BSTB
MC
SHC
C27
100nF
BSTC
USB1
TYPE-C16PIN
SHELL
GND
VBUS
SBU2
DN2
DP2
CC2
SBU1
DN1
DP1
CC1
USB_D+
USB_D-
C32
100nF
R2
5.1kΩ
R3
5.1kΩ
D2
USBLC6-2SC6
R8
5.1kΩ
USB_DET
R9
10kΩ
R11
47kΩ
ADC_VM
R12
5.1kΩ
C22
1nF
SCREW1
SCREW2
SCREW3
SCREW4
D3
D4
D5
D6
D7
D8
GHA
GLA
GHB
GLB
GHC
GLC
R17
R18
R25
R26
R36
R37
22Ω
HA
LA
HB
LB
HC
LC
Q1
Q2
Q3
Q4
Q5
Q6
R19
R20
R30
R31
R38
R39
10kΩ
MA
MB
MC
A_SHUNT+
A_SHUNT-
B_SHUNT+
B_SHUNT-
C_SHUNT+
C_SHUNT-
R22
R32
R42
50mΩ
3V3
VDD
VDDA
VREF+
R5
0Ω
C2
4.7uF
C4
100nF
C5
100nF
C7
100nF
C9
100nF
C11
1uF
C12
10nF
C15
1uF
U4
STM32G431CBT6
NRST
PG10-NRST
OSC_IN
PF0-OSC_IN
OSC_OUT
PF1-OSC_OUT
VBAT
VREF+
KEY
PC13
PC14-OSC32_IN
PC15-OSC32_OUT
ADC_VM
IC+
IC-
ADC_TP
IB-
IB+
NHC
NHB
INHA
USB_D-
USB_D+
SWDIO
SWCLK
SPI1_CSN
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8-BOOT0
PB9
PB10
PB11
PB12
PB13
PB14
PB15
IA+
IA-
SPI1_SCK
SPI1_MISO
LED
UART_TX
UART_RX
CAN_RX
CAN_TX
USB_DET
INLC
INLB
INLA
VSS
VSSA
R1
10kΩ
C3
100nF
R6
1kΩ
R7
1kΩ
LED1
LED2
U6
TCAN3414DR
CAN_TX
CAN_RX
TXD
RXD
VCC
GND
STB
CANH
CANL
SHDN/VIO
R4
120Ω
R44
10kΩ
C13
33pF
C14
33pF
X1
8MHz
OSC_IN
OSC_OUT
R43
10kΩ
H1
H2
H3
H4
H5
H7
PZ254V-11-03P
PZ254V-11-05P
PZ254V-11-01P
SWCLK
SWDIO
UART_TX
UART_RX
SPI1_CSN
SPI1_SCK
SPI1_MISO
R23
43kΩ
R21
1.5kΩ
R24
15kΩ
C29
56pF
R28
43kΩ
R27
1.5kΩ
R29
15kΩ
C30
56pF
R34
43kΩ
R33
1.5kΩ
R35
15kΩ
C31
56pF
R40
33kΩ
R41
C28
100nF
ADC_TP
TP1
TP2
TP3
TP4
TP5
TP6
TP7
TP8
TP9
TP10
TP11
TP12
原理图
Schematic1
创建日期
2026-09-04
更新日期
2026-09-09
板子
v1
图页
P1
绘制
审阅
BLDC_control
版本
V1.0
尺寸
A4
页 1 共 1
嘉立创EDA